

GRADE 11 BIOLOGY - 8-MARK "CORE OUTLETS"

One page per probable long question (Q1-Q16). Each page = ALL 8 answer points + the figure to draw + the exam gotcha. Rapid-fire recall for the night before.

USE: cover the right side, read the left, then say the 8 points out loud. One page = one question = one attempt.

Q1 CORE OUTLETS - Prokaryote vs Eukaryotic Cell (8 marks)

Draw: 1 PROKARYOTE + 1 EUKARYOTE cell side-by-side (label every part). 8 points below = full marks.

PROKARYOTE (BACTERIUM)

1. Smallest living cells, 0.2-10 µm; Kingdom Monera
2. NUCLEOID (genophore): circular DNA, NO histone, no membrane
3. NO membrane-bound organelles (no mito/Golgi/ER/plastid)
4. Ribosomes 70S = 30S + 50S (S-values NOT additive)
5. Wall = MUREIN (peptidoglycan: NAG+NAM); absent in Mycoplasma
6. Respiration on MESOSOME folds; plasmids + gas vacuoles
7. Divide by BINARY FISSION (no mitosis/meiosis)
8. Ex: Vibrio, Diplococcus, Bacillus, Streptococcus, E. coli

GUT: murein + 70S + circular DNA no histone = THE 3 markers

Mycoplasma = smallest LIVING cell (0.1-0.3 µm, no wall)

Flagellum T.S. = 9 peripheral granules (no 9+2)

Mesosome = respiration site (mitochondria-equivalent)

No true sex - recombination = CTC only

Gas vacuole = buoyancy; reserve = glycogen/fat

EUKARYOTE (PLANT / ANIMAL)

1. Larger cells, 10-100 µm; nucleus + full organelle suite
2. TRUE nucleus: double envelope + 300-1000 pores + nucleolus + chromatin
3. ALL organelles: mito, RER/SER, Golgi, lysosome, vacuole
4. Ribosomes 80S (40S+60S); 70S only inside mito + chloroplast
5. Plant wall = cellulose + plastids + big vacuole; Animal = no wall + CENTRIOLE
6. Respiration in MITOCHONDRIA; flagellum T.S. = 9+2 microtubules
7. Division = MITOSIS (somatic) / MEIOSIS (germ)
8. Ex: Amoeba, Paramecium, plants, animals, fungi

GUT: 80S + true nucleus + mito = THE 3 markers

Exception: mature mammal RBC enucleate; camel RBC = NUCLEATED

Fungus wall = chitin; flagellum 9+2 (same as animal)

Semi-autonomous organelles: mito + chloro have own DNA + 70S

Cytoskeleton + complex endomembrane system

Ostrich egg = largest cell; nerve cell = longest (90 cm)

Coenocytic exceptions: Rhizopus/Vaucheria (no walls, multi-nucleate)

CROSS-CUT 8-MARK POINTS (write BOTH columns, then these 8 lines)

- 1 Size (0.2-10 vs 10-100 µm) - 2 Nucleus (nucleoid vs true) - 3 Organelles (none vs full suite) - 4 Ribosome (70S vs 80S)
 5 Wall (murein vs cellulose/none/chitin) - 6 Respiration (mesosome vs mito) - 7 Division (fission vs mitosis/meiosis) - 8 Recombination (CTC vs meiosis)

SIMILARITY (must write!): both have plasma membrane, cytoplasm, ribosomes, DNA, reserve food - scores the "contrast" mark

MNEMONICS + DRAW THIS

Prok "M-7-C-N" = Murein, 70S, Circular DNA, No histone | Euk "8-T-M-C" = 80S, True nucleus, Mito, Cellulose (plant)

DRAW: 2 cells side-by-side, each with 6-8 labelled pointers. Title each: "Prokaryotic cell (bacterium)" / "Eukaryotic cell (animal)".

Exam gut: write the 3-marker line for EACH type first, THEN the 8 comparison points, THEN the 1 similarity line = full 8.

EXAM GOTCHAS (where marks are lost)

"All bacteria prokaryotic" = TRUE | "All protists eukaryotic" = TRUE | "Mycoplasma = smallest LIVING" = TRUE | "70S = only prokaryotes" = FALSE (also in mito/chloro)

Svedberg NOT additive: 30S+50S = 70S (not 80); 40S+60S = 80S (not 100)

SYNONYMS + ADDITIONAL OUTLETS (identity card)

Nucleoid = genophore = incipient nucleus | Mesosome = respiration fold | Cyllindrosome = P-body | Carboxysome = CO₂-fixing body

Protoplasm = living content (nucleus + cytoplasm) | Cytoplasm = jelly + organelles (excl. nucleus) | Sarcoplasm = animal cytoplasm

Cell theory "H-S-V" = Hooke (term) / Schleiden+Schwann (plant+animal) / Virchow (from cells) | Microscope "R-L-J-K-Z"

Cell theory demerits: (1) sclerenchyma/xylem dead, no protoplasm (2) Rhizopus coenocytic (3) virus = alive-ish, no cell (4) RBC + phloem enucleate

Q2 CORE OUTLETS - Eukaryotic Cell Structure & Organelles (8 marks)

Draw: 1 big ANIMAL cell + 1 close-up of 1 organelle (e.g. mitochondrion). 8 points = full marks.

NUCLEUS + MITOCHONDRIA

Nucleus: double envelope + pores + nucleolus
Mito: double membrane; cristae; matrix; 70S + 30S
Semi-autonomous: own DNA + 70S but NUCLEUS
Oxysome (F1) = ATP synthase; cristae surface = ATP synthase
Chloroplast: granum (light rxn) + stroma (dark rxn)
Both = semi-autonomous (endosymbiotic origin)
Mitochondrion count: most active cells = many
Nucleus size 5-10 μm = largest organelle; nucleolus
Chromatin = DNA + histone (euchromatin = active; heterochromatin = inactive)
Nuclear pores = 300-1000; nuclear lamina = chromatin
Germ cell = 46 chromo, 92 chromatids (after S phase)
Mature RBC = 0 nuclei (mammal); camel RBC = 1 nucleus
Glyoxysome = plant peroxisome (fatty acid $\rightarrow$ glucose)
Oxysome = F1 head of ATP synthase (Lehninger)

ER + GOLGI + LYSOSOME

RER: ribosome-studded; makes PROTEINS; corrects errors
SER: no ribosomes; makes LIPIDS; DETOX (liver)
ER = transport network; vesicles carry to Golgi
Golgi = dictyosome (plant); stacked cisternae; cis, medial, trans
Golgi: modify / package / ship proteins; make lysosomes
Lysosome: ~40 hydrolases; pH 5; "suicide bag"
Lysosome: primary $\rightarrow$ hetero $\rightarrow$ residual $\rightarrow$ autophagosome
Peroxisome: catalase kills H_2O_2 ; plant = glycogen storage
GOTCHA: lysosome = digest; peroxisome = detox
Vacuole = TONOPLE membrane = same as plasma membrane
Plant vacuole = big central; animal = small + many
Secretory vesicle = carries Golgi products to membrane
Ergastic inclusions = reserve food / secretory granules
Golgi "cis" = receiving, trans = shipping" = exocytosis

RIBOSOME + CENTRIOLE + CILIA

Ribosome 80S = 40S (large) + 60S (small); prokaryote 70S
Free ribo = cytoplasm proteins; RER ribo = secretory proteins
Polysome = many ribos on 1 mRNA; size = 23 nm
Centriole = 9 triplet microtubules (no 9+2); animal + plant
PLANT = NO centriole (uses microtubule organizing center)
Cilium / flagellum T.S. = 9+2 (9 doublets + 2 single microtubules)
9+2 = animal + plant + protozoan; 9 triplets = plant
Basal body + rootlet + kinetodesma + striated rootlets
Cilia = short, many, rhythmic; flagella = long, few, non-rhythmic
Microtubule = 13 protofilaments; microfilament = 2 protofilaments
Cytoskeleton = microtubule + microfilament + intermediate filament
Centriole "9 triplets = no central pair" vs cilium "9+2"
Bacterial flagellum = 9 granules around central rod
Cytoskeleton types: MT = 25 nm; IF = 10 nm; NF = 7 nm

SCLERE + SCALES + OTHER

Sclerosome = protein storage (seed); myelin sheath
Melanin = pigment; hemosiderin = Fe-storage
Naked-eye minimum = 100 μm ; most cells 10-100 μm
Light microscope resolution floor = 0.3 μm (Abbe's limit)
Human cells ~10¹³; RBC ~25 trillion; WBC ~100 billion
Cell = protoplasm enclosed by selective permeability membrane
Protoplasm coagulates at 60 deg C; pH slightly alkaline
Huxley: protoplasm = "physical basis of life"; protoplasm theories "P-F-H-A-B-W": Purkinje, Flemming, Huxley, Agardh, Brown, Wilson
Cytocinesis = rotation (1 vacuole) vs circulation
Demerits of cell theory: (1) sclerenchyma/xylem vessels
10 tenets of modern cell theory (name 3: correct, complete, comprehensive)
Ostrich egg = largest cell (8-12 in); nerve cell = longest
Neuroglia = non-dividing; cardiomyocytes = striated muscle

8 MARK = 1 big cell diagram (3) + 5 organelles, each structure + function (4) + 1 scale line (1)

If only space for 2 diagrams: draw (a) the whole cell and (b) 1 organelle close-up. "Semi-autonomous" = 1 mark for mito + chloro.

MNEMONIC - ORGANELLE ONE-LINERS: "N-M-G-L-R-C-V-C-P" = Never Make Green Lizards Run Cold Vibrating Caged Pigeons

Nucleus - Mito - Golgi - Lysosome - Ribosome - Centriole - Vacuole - Chloroplast - Peroxisome

DRAW: 1 animal cell (10+ labels) + 1 close-up of mito (cristae + matrix + 70S). Title both.

EXAM GOTCHAS (where marks are lost)

70S = prokaryote AND mito/chloro (NOT just prokaryote) | Plant = no centriole, no lysosome (vacuole replaces it) | Golgi "cis" = receive, trans = ship" | "9+2 = 11" for cilium

Svedberg NOT additive: 30+50=70, 40+60=80 | Mitochondrion count scales with activity (liver > muscle > RBC)

SYNONYMS + ADDITIONAL OUTLETS

Chondriosome = mitochondrion | Dictyosome = Golgi (plant) | Sclerosome = protein storage | Cyndriosome = P-body | Quantasome = light-capturing unit (230-250 Chl)

Protoplast = cell minus wall | Cytolysate = broken cell fluid | Sarcoplasm = animal cytoplasm | Protoplasm = nucleus + cytoplasm (living)

Mnemonics: "H-S-V" = cell theory | "R-L-J-K-Z" = microscope pioneers | "P-F-H-A-B-W" = protoplasm theories

EXAM GUT: name 3 tenets of cell theory (composition, arise-from-cell, function) + draw 1 organelle = full 8

Q3 CORE OUTLETS - Fluid-Mosaic Model of Plasma Membrane (8 marks)

Draw: 1 membrane cross-section - bilayer + integral + peripheral protein + cholesterol + glycoprotein. 8 points = full marks.

STRUCTURE (write components)

1. Singer & Nicolson (1972) - flexible LIPID BILAYER + embedded PROTEINS ("fluid mosaic")
2. Phospholipid bilayer: hydrophilic HEADS out + hydrophobic TAILS in (both layers face out)
3. INTEGRAL proteins: span the bilayer = channels + pumps; PERIPHERAL: on surface (adhesion)
4. Cholesterol (animal membranes): squeezed between tails - stabilizes + lowers fluidity at high temp
5. Glycoproteins / glycolipids: sugar heads on OUTER face = recognition, antigens, receptors
6. Thickness 7-8 nm; area of a cell ~4.5 m² of membrane; phospholipids flip-flop rarely

FLUID = lipids + proteins drift laterally | MOSAIC = many different proteins scattered

Asymmetry: glyco always OUTER; cholesterol mostly outer leaflet - exam point

Phospholipid % varies: high in membrane = fluid; saturated tails = rigid

Protein-to-lipid ratio differs: mitochondria = HIGH protein; myelin = mostly lipid

Membrane = "living" (selective); a dead membrane is only semipermeable

Endo/vesicle/exocytosis = membrane changes shape + area (not just diffusion)

TRANSPORT (write 3 mechanisms)

1. DIFFUSION (passive): small non-polar (O₂, CO₂) cross freely down gradient, no ATP
2. OSMOSIS: movement of WATER through semipermeable membrane; hypertonic = shrinks
3. FACILITATED diffusion: water (AQUAPORIN) + ions + glucose via CHANNEL / CARRIER proteins
4. ACTIVE transport: carrier PUMPS against gradient = ATP (Na⁺/K⁺ pump: 3 Na⁺ out, 2 K⁺ in)
5. VESICULAR: ENDOCYTOSIS (fluid / receptor / pinocytosis-phagocytosis) vs EXOCYTOSIS (secretion)
6. Permeability ladder: permeable > semipermeable > selective (differentially) permeable

Selective / differentially permeable = the LIVING membrane (Loewy & Siekevitz 1963)

Osmosis direction: low concentration → high; water potential ψ (negative = more water moves in)

Turgor = plant cell in hypotonic; flaccid = isotonic; plasmolysed = hypertonic

Glycoprotein = cell recognition ("self vs foreign") + antigen sites + receptor sites

Carrier proteins = saturated at high load (V_{max}) - exam differentiator vs channel

Membrane coagulates at 60 °C (protoplasm limit) - "living membrane" dies = permeable only

FUNCTIONS (write 5 functions)

1. SELECTIVE permeability (lets in/out chosen molecules) | 2. BARRIER (protection + compartmentalization)
3. TRANSPORT (all 5 mechanisms above) | 4. CELL RECOGNITION (glycocalyx) + adhesion
5. RECEPTOR sites (hormone/antibiotic bind) | 6. ENZYME activity (membrane-bound enzymes)
7. SHUTTLING organelle-to-organelle (ER → Golgi vesicles ride the membrane system)
8. SIGNAL transduction + junctions (tight / desmosome / gap junctions between animal cells)

Membrane potential = -70 mV (resting, Na⁺/K⁺ pump); action potential = depolarization

GOTCHA: "semipermeable = living"? NO - living membrane = SELECTIVELY permeable (chose)

SYNONYMS + EXPERIMENTS

Glycocalyx = carbohydrate coat outside membrane | Aquaporin = water channel protein

Frederick Gony (1896) = first osmotic experiment | Gorter & Grendel (1925) = LIPID BILAYER

Danielli-Davson (1935) = PMM sandwich | Singer-Nicolson (1972) = FLUID-MOSAIC (modern)

Freeze-fracture EM = sees the MOSAIC (proteins like lakes in an ice sheet)

Permeability ladder order: PERMEABLE (wall) > SEMI (parchment) > SELECTIVE (living membrane)

Synonyms: phospholipid = glycerophospholipid; cholesterol = sterol; glycolipid = cerebroside

GOTCHA: plant cell has MEMBRANE + WALL (cellulose); wall = PERMEABLE, membrane =

8 MARK = 1 diagram (2) + 4 components (4) + 2 functions (2). Draw bilayer + 5 labelled structures

First line of answer: "Fluid = can move laterally; Mosaic = different proteins scattered" = the question itself. State "Singer & Nicolson 1972".

MNEMONICS + EXAM GUTS

"FLUID-MOSAIC" = 2 words = write both in line 1. "Na⁺/K⁺ pump: 3 out, 2 in, 1 ATP" = the active-transport mark

Transports: D-F-A-V-E = Diffusion, Facilitated, Active, VESicular, ENDocytosis (5 types, know 3-4)

EXAM GUT: osmosis = WATER movement (NOT solute); plasmolysis = in HYPERTONIC; turgor = HYPOSTONIC

Q4 CORE OUTLETS - Bacterial Cell Structure + Bacteria vs Archaea (8 marks)

Draw: 1 labelled BACTERIUM (rod or coccus) with all 9-10 parts. 8 points = full marks.

STRUCTURE - THE 10 LABELS

1. PLASMA MEMBRANE (phospholipid + protein; transport + respiration site)
 2. MESOSOME = infolding of membrane = respiration (mitochondria-equivalent)
 3. NUCLEOID (genophore): circular DNA + RNA + protein, NO histone, no membrane
 4. PLASMID: small circular DNA - antibiotic-resistance + gene-transfer vehicle (F-plasmid)
 5. CELL WALL = MUREIN: N-acetylglucosamine + N-acetylmuramic acid (NOT cellulose)
 6. CAPSULE (slime layer): anti-phagocytic (Klebsiella, S. pneumoniae); virulence factor
 7. FLAGELLUM (flagellin protein; T.S. = 9 granules + central rod); cilia-like = PILI (conjugation)
 8. GAS VACUOLE (buoyancy in aquatic types; NO true vacuole) | reserve = glycogen / fat
 9. RIBOSOME 70S (30S + 50S); 10. ENDOSPORE (some; survival NOT reproduction)
- GUT: murein + 70S + circular DNA no histone = THE 3 exam markers**
- Size 0.2-10 µm; Mycoplasma = smallest LIVING (0.1-0.3 µm, no wall, pleomorphic)
Mycoplasma = "Joker of plant kingdom" + penicillin-resistant (no wall = no target)

BACTERIA vs ARCHAEA (the table)

Wall: murein (bact) vs PSEUDOMUREIN (arch) - first row of the table
Membrane lipids: ESTER bonds (bact) vs ETHER bonds (arch) - signature of archaea
RNA polymerase: single-subunit (bact) vs multi-subunit eukaryote-like (arch)
Methanogens: ARCHAEA only (produce CH₄ in anaerobic mud, rumen, waste)
Extremophiles: ARCHAEA signature (thermophiles / halophiles / acidophiles)
Antibiotic sensitivity: bact = yes; arch = generally NO
Habitat: bact = everywhere (ubiquitous); arch = mostly extreme (hot springs, salt lakes)

"All bacteria are prokaryotic" = TRUE | "All protists are eukaryotic" = TRUE

Archaea = 3rd domain (Woese 1977); both prokaryotes = NO true nucleus, circular DNA
Methanogenesis = H₂ + CO₂ → CH₄ (archaea); acetate → CH₄ + CO₂
"Blue-green algae" = cyanobacteria = BACTERIA, not algae (exam trap)
Archaea membrane = ISOPRENE + ether; bacteria = fatty acid + ester

SHAPE (Cohn's classification) - write 5-6 shapes

COCCHI (round): diplococcus / streptococcus / staphylococcus / tetrads / sarcina
BACILLI (rod): diplo- / strepto- / single; with or without endospore
VIBRIO (comma): V. cholerae (cholera) | SPIRILLUM (rigid spiral): S. volutans = longest
SPIROCHAETE (flexible spiral): T. pallidum = syphilis | SPHEROIDAL = cocci in chains
CYANOBACTERIA (filamentous, "blue-green"): Nostoc, Anabaena (heterocyst), Oscillatoria

MYCOPLASMA = pleomorphic / irregular (no wall = shape-free) - the "joker"

Arrangement: "D-S-T" = Diplo (2) / Strepto (chain) / Tetrads + Staphylo (4 / bunch)

SYNONYMS + WALL TYPES (Gram stain)

Murein = mucopeptide = peptidoglycan | Nucleoid = genophore | Cylindrosome = P-body
GRAM +: thick murein 20-80 nm + teichoic acid → crystal violet = BLUE
GRAM -: thin murein 5-10 nm + outer LPS membrane → safranin = PINK
ACID-FAST: Mycobacteria (waxy mycolic acid); Ziehl-Neelsen = red; TB + leprosy
NO-WALL: Mycoplasma (pleomorphic + osmotic-lysis risk + penicillin-resistant)
LPS = ENDOTOXIN of Gram- (septic-shock risk); penicillin attacks murein synthesis

GUT: "G-P thick, G-N thin + outer" + "Murein = NAG + NAM (NOT cellulose)"

8 MARK = 1 diagram (3) + 5 structure points (4) + Bacteria-vs-Archaea 1-row (1)

The 5 structure points: nucleoid + 70S + murein + mesosome + plasmid / gas vacuole. ONE line each.

MNEMONICS + EXAM GUTS

"M-7-C-N" = Murein, 70S, Circular DNA, No histone | "G-P thick, G-N thin + LPS outer, acid-fast waxy, no-wall mycoplasma"

Shape "C-D-S-T" = Coccus → Diplo → Strepto → Staphylo → Tetrads | Archae signature = "ester / ether / methanogen / extremophile"

GUT: "Bacteria = smallest living" = FALSE (Mycoplasma yes; virus no - not living). Endospore = survival NOT reproduction (1 → 1 → 1)

Q5 CORE OUTLETS - Nutrition in Bacteria (Classified) (8 marks)

Draw: 1 classification tree (3 keys → 4 types) + 1 O₂-requirement row. 8 points = full marks.

3 KEYS "E-C-E" + 4 TYPES

ENERGY (light / chemo) · CARBON (CO₂ / organic) · DONOR (inorganic / organic)
 PHOTOAUTOTROPH: light + CO₂; e.g. Cyanobacteria, green / purple S bacteria
 CHEMOLITHO: inorganic oxidation + CO₂; nitrifiers (Nitrosomonas / Nitrobacter), S (Thiobacillus)
 CHEMOORGANO: organic energy + C; MOST types - E. coli, Staphylococcus, Lactobacillus
 PHOTOHETERO: light + organic C; Rhodospirillum, Chloroflexus (non-sulfur)
"PHL / CTL / CHO / PHL": photo-litho-AUTO / chemo-litho-AUTO / chemo-organo-HETERO
 Autotroph = makes own C from CO₂ (like a plant); heterotroph = eats ready organic C
 Litho = inorganic donor; organo = organic donor. 2 × 2 = 4 common types (+ photo-hetero)
 Photoautotrophs split into OXYGENIC (cyanobacteria, O₂ out) vs ANOXYGENIC (S-bact, S out)
GUT: split H₂O = O₂ released (cyanobacteria); split H₂S = NO O₂, S deposits (green/purple)
 Purple non-sulfur + green non-sulfur = photoHETERO (organic donor, anoxygenic)

CHEMOLITHOTROPHS - THE "SOIL CLEANERS"

NITRIFYING: NH₃ → NO₂⁻ (Nitrosomonas) → NO₃⁻ (Nitrobacter) - 2 steps, AEROBIC
 SULFUR: H₂S / S / S²⁻ → SO₄²⁻ (Thiobacillus, Beggiatoa) = sulfur cycle
 IRON: Fe²⁺ → Fe³⁺ (Ferrobacillus); HYDROGEN: H₂ → H₂O (Rickettsia)
 METHANOGENS (ARCHAEA): H₂ + CO₂ → CH₄ = anaerobic, NOT bacteria
 ALL chemoautotrophs: inorganic energy + CO₂ carbon + aerobes (EXCEPT H₂ + methanogen)
GUT: "nitrification = NH₃ → NO₃⁻; denitrification = NO₃⁻ → N₂; fixation = N₂ → NH₃"
 Nitrogenase = O₂-SENSITIVE → cyanobacteria fix N₂ in HETEROCYSTS (oxygen-free cell)
 Rhizobium = symbiotic N₂-fixer (legume nodules); Azotobacter / Clostridium = free-living
 Heterocyst = thick wall, no PSII, O₂-free - the exam "special" cell
 "Blue-green algae" = cyanobacteria = BACTERIA (not algae) + fix N₂ + O₂-evolving

CHEMOORGANOTROPHS - 3 LIFESTYLES + GLUCOSE FATES

SAPROPHYTE (decomposer): feeds on DEAD organic = Lactobacillus, soil bacteria, mineralizer
 PARASITE: feeds on LIVING host = Staphylococcus, Salmonella, Mycobacterium (disease)
 SYMBIONT (mutual): both benefit = Rhizobium (fixes N₂ for the legume)
 Glucose fates: AEROBIC respiration (O₂ → CO₂ + H₂O, ~30-38 ATP)
 ANAEROBIC respiration (NO₃⁻ / SO₄²⁻ acceptor, moderate ATP) | FERMENTATION (organic acid)
 Lactic (Lactobacillus, muscle) + Ethanol (yeast) = the 2 fermentation products
GUT: "denitrification = NO₃⁻ → N₂ (Clostridium, Pseudomonas) - LACKS N from soil"

O₂ REQUIREMENT - 6 TYPES (mnemonic "O-OF-M-A")

OBLIGATE AEROBES (need O₂): Mycobacterium, Pseudomonas
 OBLIGATE ANAEROBES (O₂ KILLS): Clostridium, Methanogens
 FACULTATIVE ANAEROBES (prefer O₂, can ferment): E. coli, Staphylococcus
 MICROAEROPHILES (little O₂ only): Helicobacter, Campylobacter
 AEROTOLERANT (no O₂ needed, not harmed): Lactobacillus, Streptococcus
 PHOTOAEROTOLERANT = phototroph + O₂ OK (green/purple non-sulfur)
"Obligate = forced; Facultative = can switch; Micro = small dose; Aero-tolerant = not harmed"

8 MARK = 1 tree (3) + 4 types, each source + example (4) + 1 O₂-requirement line (1)

Draw the 3 keys (E / C / D) → 4 types + name ONE example for EACH type = the "identifiable" mark.

MNEMONICS + EXAM GUTS

"E-C-E" = Energy / Carbon / Electron-donor | "PHL / CTL / CHO" = the 4 types by root-words
 N-cycle "F-N-D" = Fixation (N₂ → NH₃) → Nitrification (NH₃ → NO₃⁻) → Denitrification (NO₃⁻ → N₂)
GUT: Rhizobium = the ONE chemo-hetero that FIXES N₂ (symbiont) | Methanogens = ARCHAEA not bacteria | "O₂ KILLS it" = obligate anaerobe

Q6 CORE OUTLETS - Bacterial Reproduction + Genetic Recombination (8 marks)

Draw: 1 binary-fission sequence (3 frames) + 1 CTC arrow diagram. 8 points = full marks.

ASEXUAL - BINARY FISSION (main)

1. DNA replicates → cell elongates → septum forms → 2 IDENTICAL daughter cells (clones)
2. Generation time: *E. coli* ~20 min; exponential 1 → 2 → 4 → 8 (under ideal conditions)
3. Bacteria divide by BINARY FISSION only (no mitosis / meiosis - prokaryotes)
4. Some by BUD (yeast-like, e.g. *Hyphomicrobium*); few by FRAGMENTATION / multiple fission
5. 1 PARENT → 2 DAUGHTER = population doubles (geometric growth)

GUT: "binary fission = 1 → 2 clones" - the asexual mark

ENDOSPORE = SURVIVAL (NOT reproduction)

1. Formed by *Bacillus* + *Clostridium* in harsh (dry / heat / nutrient-poor) conditions
2. Heat + acid + chemical-proof; stays dormant years; germinates when water + food return
3. Spore vs vegetative cell: spore = dehydrated core + cortex + exospore, metabolically inert
4. NOT used to increase numbers (1 spore = 1 new cell when it germinates, not 2)

EXAM TRAP: "endospore = reproduction" is WRONG - it is 1 cell → 1 spore → 1 cell (survival)

GENETIC RECOMBINATION "CTC" (NO true sex)

1. CONJUGATION: F+ donates F-plasmid via SEX PILUS → F- becomes F+ (plasmid transfer, no DNA)
2. TRANSFORMATION: cell takes up FREE (naked) DNA from the environment (Griffith 1928)
3. TRANSDUCTION: BACTERIOPHAGE carries donor DNA into a new host during infection
4. Recombination shuffles genes WITHOUT making more cells = variability + resistance transfer
5. Plasmid + transposon = "jumping genes" + antibiotic-resistance spread (exam point)
6. No syngamy, no meiosis, no true sex in bacteria - recombination only

MNEMONIC "CTC" = Conjugation (pilus) / Transformation (naked DNA) / Transduction (phage)

SYNONYMS + GENE-TOOL LINKS

F-plasmid = fertility factor | "F+ donor / F- recipient" | sex pilus = conjugation bridge
 Transformation = basis of GENE ENGINEERING (recombinant DNA uptake)
 Transduction types: generalized (random DNA) vs specialized (near prophage genes)
 "Recombinant" = cell carrying new allele; Hfr = high-frequency recombination strain
 Mesosome may anchor the DNA replication origin (exam link to structure)

GUT: CTC = the ONLY genetic mixing in prokaryotes - memorize the 3 + one line each

8 MARK = 1 fission diagram (2) + 3 recombination types (4) + "no true sex" + significance (2)

Significance: recombination gives variability + antibiotic-resistance transfer via plasmids & transposons.

MNEMONICS + EXAM GUTS

Reproduction "B-E-C" = Binary fission (asexual) / Endospore (survival, NOT repro) / CTC (recombinants)

"CTC" one-liners: Conjugation = PILUS + F-plasmid | Transformation = NAKED DNA from soil | Transduction = PHAGE as courier

GUT: "no true sex in bacteria" = asexual binary fission + 3 CTC recombination events (NOT syngamy / meiosis)

4-mark mini: 3 CTC types + 1-line each + "no true sex" | 4.8 = add a labelled fission + a conjugation pair

Draw: 3-frame binary fission (DNA copies → septum → 2 clones) + 2 bacteria linked by a pilus = conjugation

Q7 CORE OUTLETS - 4 Classes of Protozoa (by locomotion) (8 marks)

Draw: 1 protozoan (e.g. Amoeba or Paramecium) labelled. 4 classes, each: locomotion + example + 1 feature = full marks.

1. SARCODINA (Rhizopoda / Amoeboid)

- LOCOMOTION: pseudopodia ("false feet", cytoplasmic streaming) - no rigid covering
- Nutrition: PHAGOCYTOSIS (engulf → food vacuole + lysosome); forms CYSTS (resting spores)
- Ex: Amoeba proteus, Entamoeba histolytica (dysentery), Arcella, Polyoma
- IDENTIFY: "irregular crawling + pseudopod" = Sarcodina | Synonym: Rhizopoda = Sarcodina**
- Reproduction: binary fission + (some) conjugation; 1 or many nuclei

2. MASTIGOPHORA (Flagellates)

- LOCOMOTION: FLAGELLA (whip-like, 1 to many); flexible PELICLE (no rigid wall)
- Mixotrophic = photosynthesis + heterotrophic (Euglena); red "eyespot" (stigma) senses light
- Ex: Euglena, Trypanosoma (sleeping sickness / Chagas; flagellum + undulipodium), Giardia
- IDENTIFY: "whip-like flagellum + plicella" = Mastigophora | Giardia = "flagellate with 2 nuclei"**
- Dinoflagellates: 2 flagella (one transverse = spin); bioluminescent; some form red tides

3. CILIOPHORA (Infusoria / Ciliates)

- LOCOMOTION: CILIA (short, many, rhythmic, body-covering) = most complex protozoa
- TWO nuclei: MACRO (somatic) + MICRO (germ); food → oral groove / cytopharynx; 2 contractile vacuoles
- Ex: Paramecium caudatum, Opalina, Stentor, Vorticella, Balantidium (see Q12-13)
- IDENTIFY: "many cilia + 2 nuclei" = Ciliophora | Synonym: Infusoria = Ciliates**
- Reproduction: binary fission + CONJUGATION (reciprocal haploid exchange) + encystation

4. SPOROZOA (Apicomplexa)

- LOCOMOTION: NONE in the adult = obligate PARASITE; spore dispersal; apical complex (rhoptries)
- Digenetic = 2 hosts; sexual + asexual ALTERNATION; spore-forming (sporogonium)
- Ex: Plasmodium (malaria), Toxoplasma gondii (cat-spread, dangerous in pregnancy), Cryptosporidium
- IDENTIFY: "parasite + spore + no cilia / flagella" = Sporozoa | Asexual = schizogony / sporogony**
- Reduced organelles; see Q10-11 for full Plasmodium life cycle

NUTRITION TYPES (across protozoa)

HOLOZOID (ingest - ciliates) / OSMOTROPHIC (absorb - slime mould, parasitic) / PHOTOSYNTHETIC (algae)
 Protista = the eukaryote "residual" kingdom: not plant / animal / fungus (Haeckel 1869)
 "The 2 algal divisions" often listed = Euglenophyta / Chrysophyta (diatom siliceous test) - CHLOROPHYTES

SYNONYM BANK (1/1/1)

Rhizopoda = Sarcodina (Amoeboid) | Mastigophora = Flagellates | Ciliophora = Infusoria / Ciliates
 Protozoa = protozoans = primitive animals (loose) | Plicella = flexible pellicle | Stigma = eyespot
 Undulipodium = flagellum edge membrane (Trypanosoma) | Rhoptries = apical invasion organelles

THE MASTER KEY "P-F-C-N" = the class = the locomotion organ (memorize + 1 example + 1 feature each)

P = Pseudopod (Sarcodina) | F = Flagellum (Mastigophora) | C = Cilia (Ciliophora) | N = None (Sporozoa, parasitic)

EXAM GUTS + 8-MARK PATTERN

8 MARK = 1 protozoan diagram (2) + 4 classes, each locomotion + example + 1 feature (4) + 1 algal-division + lifestyle line (2)

"Ciliophora's TWO nuclei" is its signature | "Sporozoa = no locomotion + spore + apical complex" | write "P-F-C-N" first, then 1 example + 1 extra per class

Boundary: the exact "6 classes" name-list varies by book - 4 animal + 2 algal divisions; verify YOUR book's 6 before the exam

Q8 CORE OUTLETS - Structure, Nutrition & Reproduction of EUGLENA (8 marks)

Draw: 1 labelled Euglena (slipper-shaped) - 6 parts. 8 points = full marks.

IDENTITY + STRUCTURE (the 6 labels)

1. Identity: *Euglena gracilis* = a MASTIGOPHORAN; slipper-shaped, 60-70 µm; stagnant plant-animal hybrid
2. PELICLE: thin flexible covering under the membrane (no rigid wall) - gives shape + allows movement
3. FLAGELLUM: 1-2 long whip-like flagella = locomotion (swimming)
4. CHLOROPLAST + EYESPOT (stigma): photosynthesis; the red stigma SENSES light so it swims towards it
5. NUCLEUS + CONTRACTILE VACUOLE (pulsatile): one true nucleus; the CV expels excess water
6. PARAMYLON (red) + pyrenoid (starch granule storage) inside the chloroplast; reserves = para

LABELS to draw: pellicle, flagellum, eyespot, chloroplast (+pyrenoid), nucleus, contractile vacuole

- Spirochaete (red) + pyrenoid (starch granule storage) inside the chloroplast; reserves = para
- Spiro-excretion by diffusion through pellicle (no respiratory organ)
- Some forms a resting CYST (epilimnium) when conditions turn bad

NUTRITION + REPRODUCTION + SIGNIFICANCE

NUTRITION = MIXOTROPHY (the key word):

- IN LIGHT: PHOTOSYNTHETIC (plant-like autotrophy - uses its chloroplasts)

- IN DARK: HETEROTROPHIC (absorbs dissolved organics) - "mixotrophic" scores the whole mark

REPRODUCTION: asexual by LONGITUDINAL binary fission; some form a resistant cyst (hypoc

Eyespot (stigma) = light sensor, NOT an eye - the exam "do not call it an eye" trap

SIGNIFICANCE: model for studying the plant-animal boundary; used in WASTEWATER TREATMENT

- Shows the "residual" nature of Protista (one cell with BOTH cell types' machinery)

Euglena = a MASTIGOPHORAN that is also photosynthetic - the "both worlds" protzoan

SYNONYMS (1/1/1) + EXAM GOTCHAS

Euglena = slipper animal (common) | Plicella = pellicle (synonym) | Stigma = eyespot = photosensor

Paramylon = "animal starch" storage | Pyrenoid = CO₂ concentrating body in chloroplast

"*Euglena* = algae?" - NO, it is a protozoan (Mastigophora) that happens to photosynthesize

"*Euglena* = animal?" - it is a protozoan; it swims, senses light, ingests in dark - a hybrid

GOTCHA: "eyespot = eye" is WRONG - it is a light detector that shows the way to light

DIFFERENCE FROM OTHER MASTIGOPHORANS

Euglena = mixotrophic + photosynthetic + plicella + eyespot (free-living, benign)

Trypanosoma = parasitic + no chloroplast + flagellum + undulating membrane (disease)

Giardia = 2 nuclei + 8 flagella + no chloroplast (diarrhoea) | *Leishmania* = parasitic + epimastigote

Euglena is the ONLY MASTIGOPHORAN that is photosynthetic (the green one)

IDENTIFY: "slipper + flagellum + red eyespot + chloroplast" = *Euglena*

8 MARK = 1 *Euglena* diagram (3) + 5 structure parts (3) + nutrition + reproduction lines (2)

Write "MIXOTROPHIC" first - it earns the whole nutrition mark: photosynthetic in light, heterotrophic in dark.

MNEMONICS + EXAM GUTS

Euglena structure "P-F-E-C-N-V" = Pellicle / Flagellum / Eyespot / Chloroplast / Nucleus / Contractile vacuole

Mastigophoran exam trio: (1) flagellum = locomotion (2) plicella = flexible covering (3) some = photosynthetic

"P-F-C-N" = the protozoan class key - *Euglena* is under "F" (flagellum) with photosynthetic ability

EXAM GUT: "mixotrophic" = the ONE word for *Euglena*'s nutrition - say "plant in light, animal in dark"

Eyespot = red light sensor (NOT an eye) | Draw: slipper + flagellum + red eyespot + green chloroplast + nucleus + CV

4.8-mark extension: add the pyrenoid + paramylon + "bridges the plant-animal boundary" line for the extra 0.8

Q9 CORE OUTLETS - Beneficial & Harmful Importance of Protozoa (8 marks)

Draw: 1 protozoan (Amoeba or Trypanosoma). 4 beneficial + 4 harmful (disease + protozoan + route) = full marks.

BENEFICIAL USES (write 4)

1. O₂ + FOOD BASE: photosynthetic protozoa (Euglena, diatoms, dinoflagellates) = major O₂
 2. FOOD FOR ANIMALS: paramecia & infusoria feed larvae of fishes, frogs & insects (live food)
 3. WATER TREATMENT: ciliates in sewage / wastewater remove bacteria & organics; some = BIOFILM
 4. LAB / RESEARCH: Paramecium & Amoeba are classic model organisms; used in protist reproduction
 5. NUTRIENT RECYCLE: protozoan grazing on bacteria = "protozoan loop" (recycles N + C back to plants)
 6. SOME symbiont protozoa help digest cellulose in termite guts (protosymbiosis with bacteria)
- "Beneficial protozoa = food, oxygen, water-filtering + lab study models" - one line each**
- Dinoflagellates also run "red tides" (harmful) - double-edged

HARMFUL DISEASES (write 4: protozoan + disease + route)

1. AMOEBIC DYSENTERY: Entamoeba histolytica (liver abscess) - FECAL-ORAL (contaminated food/water)
 2. SLEEPING SICKNESS (African): Trypanosoma brucei - TSETSE fly (Glossina); CHAGAS: T. cruzi - triatomine bug
 3. GIARDIASIS (diarrhoea): Giardia lamblia - water-borne | TRICHOMONIASIS: Trichomonas vaginalis - sexual
 4. MALARIA: Plasmodium (falciparum / vivax / ovale / malariae / knowlesi) - female ANOPHELES mosquito
 5. TOXOPLASMOSIS: Toxoplasma gondii - spread by CATS; dangerous in pregnancy (miscarriage, stillbirth)
- "The deadliest human protozoan = Plasmodium (malaria)" - name it first for the biggest mark**
- Leishmania = leishmaniasis (sand-fly vector); Naegleria fowleri = brain-eating amoeba (rare, via swimming pool)
- PAIR each disease with its protozoan + ROUTE (exam wants all three, not just the disease name)

THE 4-DISEASE MARK-LIST (memorize + say route)

Plasmodium = MALARIA | Trypanosoma = SLEEPING SICKNESS / CHAGAS | Entamoeba = DYSENTERY
 Toxoplasma = TOXO (cats) | Trichomonas = TRICH (venereal) | Leishmania = LEISH (sand-fly)
 Route: protozoan + VEHICLE (fly / cat / water / fly / fecal-oral / mosquito)

"Malaria = the biggest protozoan disease" - if you can only name one, name it

SYNONYMS + EXAM GUTS

Sporozoa = Apicomplexa (parasite class) | "Amoebiasis" = "amoebic dysentery" (Entamoeba)
 Phytoplankton = photosynthetic protozoa / algae that form the aquatic food base + O₂
 Bio-indicator = species whose presence shows water quality (protozoan index) - "protozoan index"
 Vector vs host: mosquito = vector (carries); human = host (harbours); tsetse = vector; termite = host
GUT: "beneficial = food / O₂ / filter / lab" | "harmful = 4 named diseases + route" - one line each

8 MARK = 2 protozoan diagrams (2) + 4 beneficial (3) + 4 harmful diseases (3)

Pair EACH disease with its protozoan + its route - "name only the disease" loses the second mark each time.

MNEMONICS + EXAM GUTS

Harmful "P-T-E-G-M" = Plasmodium (malaria) / Trypanosoma (sleeping) / Entamoeba (dysentery) / Giardia (diarrhoea) / Toxo (+ Trichomo / Leish)
 Beneficial "F-O-F-L" = Food (larvae) / O₂ (phytoplankton) / Filter (water) / Lab (model) - one line each
 Protozoan loop = protozoa eat bacteria, release nutrients back to plants - the "recycler" role
EXAM GUT: "the deadliest = Plasmodium (malaria)" + "beneficial = O₂ / food / filter / lab" - those 2 lines are the skeleton
 4-mark mini: 2 diseases + 2 benefits | 4.8 = all 4 harmful + 4 beneficial + 1 vector / host distinction
 Draw: 1 protozoan + a small "arrow map" linking protozoan → disease → route (visual, wins +1 mark)

Q10 CORE OUTLETS - COMPLETE LIFE CYCLE OF PLASMODIUM (malaria) (8 marks)

Draw: the 7-BOX cycle - 4 boxes = HUMAN (blue), 3 boxes = MOSQUITO (red). 8 points = full marks.

HUMAN PHASE (asexual) - 4 steps

1. BITE: female Anopheles injects CRESCENTIC SPOROZITES into the blood
 2. LIVER (hepatic / pre-erythrocytic): sporozoites multiply by SCHIZOGONY → MEROZOITES (liver stage)
 3. BLOOD (erythrocytic): merozoites invade RBCs → TROPHOZOITE → schizont → RBC bursts
 4. GAMETOCYTES form: **ONLY the form the mosquito can ingest (round macrogamete / c)**
- vivax / ovale: HYPNOZOITES stay dormant in liver → RELAPSE later (tertian relapsing malaria)
- "S-T-S-G" = 4 forms: Sporozoite / Trophozoite / Schizont / Gametocyte

MOSQUITO PHASE (sexual) - 3 steps

5. Gut: microgamete EXFLAGELLATES + fertilizes macrogamete → ZYGOTE
 6. Zygote → motile OOKINETE → OOCYST (SPOROGONY) → SPOROZITES migrate to SALIVARY GLANDS
 7. Now-infective mosquito bites a NEW human → cycle repeats (DIGENETIC = 2 hosts + 2 kinds of generation)
- "Sporozoite infects the HUMAN; gametocyte infects the MOSQUITO" = the classic exam line!**
- falciparum microgametocyte = BANAAN / CRESCent shape; vivax = ROUND
- Sexual generation = GAMETOGENY (in mosquito); asexual = SCHIZOGONY + SPOROGONY

MNEMONIC - the 7 boxes in order: "BITE → LIVER → BLOOD → GAMETO → ZYGOTE → SALIVA → BITE"

Human = L + B (liver + blood) | Mosquito = Z + S (zygote + salivary) | Colour-code HUMAN blue / MOSQUITO red when you draw

CLASSIFICATION (write it - the hidden mark)

Domain Eukarya · Protozoa (Sporozoa / Apicomplexa) · Phylum Apicomplexa · Class Coccidia · Order Haemosporidia · Family Plasmodiidae · Genus Plasmodium · Species P. falciparum / vivax / malariae / ovale
Vector: female ANOPHELES only (daytime / dusk) | Host: humans (vivax/ovale = also monkeys)
"P-L-F" = Periodicity: falciparum IRREGULAR; vivax/ovale = TERTIAN 48h; malariae = QUARTAN 72h
4 FORMS: crescentic SPOROZOITE / ring TROPHOZOITE / rosette Schizont (malariae) / gametocyte
Merozoite = liver-burst product; gametocyte = mosquito-eatable; ookinete = mobile zygote
falciparum = most dangerous (cerebral malaria, NO rosette); vivax = Schuffner's dots + relapse

SYMPTOMS + CONTROL (write 4 control lines)

Symptoms: chills → fever → sweating (tertian "QTTK" = 48h); anaemia (RBC destruction); splenomegaly
Cerebral malaria (falciparum): brain infection = highest death rate; blackwater urine (haemoglobinuria)
CONTROL 1: drain stagnant water + cover water stores (break mosquito breeding)
CONTROL 2: DDT spraying + bed-nets (vector control - female Anopheles is the ONLY target)
CONTROL 3: antimalarial drugs (chloroquine - falciparum; ACTs; PEP = primaquine kills hypnozoites)
CONTROL 4: G6PD test before primaquine; vaccine trials; screen the patient + blood donors
GUT: "drain + DDT + nets + drugs" = the 4 standard control lines examiners expect

8 MARK = 1 7-box life cycle diagram (4) + 4 human-phase steps (2) + 4 mosquito + digenetic line (2)

Draw the 7 boxes in 2 rows (4 blue human + 3 red mosquito), arrows showing: human RBC → mosquito gut → salivary → new human

MNEMONICS + EXAM GUTS

"BITE-LIVER-BLOOD-GAMETO-ZYGOTE-SALIVA-BITE" = 7 steps | "S-T-S-G" = the 4 human-phase forms
Fever "QTTK": Q = quartan (malariae 72h) / TT = tertian (vivax + ovale 48h) / K = quotidian (knowlesi 24h) / falciparum = irregular
"Digenetic" = 2 hosts + 2 kinds of generation (asexual in man, sexual in mosquito) - write it as a signature line
EXAM GUT: "sporozoite infects the human; gametocyte infects the mosquito" - that one line wins the whole diagram mark
4-mark mini: human 4 steps + mosquito 3 steps | 4.8 = add 1 species (falciparum crescent + irregular fever) + 1 control line
"Exflagellation" = microgamete gets 3 flagella in the mosquito gut - a 0.8-mark bonus detail

Q11 CORE OUTLETS - DIFFERENTIATE THE 5 SPECIES OF PLASMODIUM + CONTROL (8 marks)

Write: 5-row species table + 1 fever-pattern line + 4 control measures = full marks.

THE 5 SPECIES - THE TABLE (write 6 columns, 5 rows)

SPECIES | FEVER PERIOD | RBC INVADED | SIGNATURE MARK | HYPNOZOITE?

P. falciparum | IRREGULAR / malignant | ALL ages of RBC | CRESCENT / BANANA microgametocyte | NO = 1

P. vivax | TERTIAN 48 h (benign) | OXYGENATED RBC | SCHUFFNER'S DOTS + ameoboid trophozoite | YES (relapse) = 1

P. ovale | TERTIAN 48 h | YOUNG RBC only | POINTED / deformed RBC + dots | YES (relapse) = 1

P. malariae | QUARTAN 72 h | OLD RBC | ROSETTE / DAISY-PETAL schizont + band trophozoite | NO = 1

P. knowlesi | QUOTIDIAN 24 h | — | ZOONOTIC (macaque → man), enlarges fast | — = 1

GUT: each row = 1 mark; falciparum row (diesis of crescent + "no tertian regularity") = the hardest mark

RBC rule of thumb: falciparum = all RBC + crescent; vivax = oxygenated + dots; ovale = YOUNG only + pointed; malariae = OLD + rosette

"Rosette / daisy" = malariae's ring of 8 schizonts; "Schuffner's dots" = red stippling in vivax-infected RBCs

Diagnosis: THICK film = detection, THIN film = species ID + parasitemia; G6PD before primaquine

Knowlesi: "K-24" = macaque-borne, 24-hour quotidian cycle - fast-growing, dangerous in pregnancy

Fever: chills → fever → sweating → remission; periodicity = burst cycle length (48h vs 72h vs 24h)

4 CONTROL MEASURES (write exactly these)

1. DRAIN / remove stagnant water + cover water containers (break larval habitat)
2. DDT / insecticide spraying + INSECT-TREATED BED-NETS (vector = female Anopheles ONLY)
3. ANTIMALARIALS: chloroquine (falciparum, not vivax), ACTs; PEP = primaquine kills liver hyp
4. SCREEN + VACCINE trials; screen blood donors; isolate sick + fever RDT (Rapid Diagnostic Test)

"Drain + DDT + nets + drugs" = the 4 standard control lines examiners expect

Prevention = "P-P-P" = Prophylaxis (drugs) + Protection (nets + DDT) + Puddle-draining

FEVER PATTERN "QTTK" + RELAPSE RULE

Q = QUARTAN (72h) = P. malariae | TT = TERTIAN (48h) = P. vivax + P. ovale | K = 24h = P. knowlesi

RELAPSE RULE: ONLY vivax + ovale relapse (liver HYPNOZOITES wake later) - their exam idea

Fever type = duration of the erythrocytic cycle × 3 - the "why" line (48h × 3 = tertian = every 48h)

Malignant tertian = falciparum; benign tertian = vivax / ovale - the "benign" mark word

Chills → Fever → Sweating = 3-part paroxysm; cerebral malaria = falciparum-specific

"Anox = nocturnal / night" (Anopheles bites at dusk / dawn; vivax = daytime; ovale = also relapse)

8 MARK = species-table (5) + fever-pattern (1) + 4 control (2)

Draw the 5-row table with ALL 6 columns; colour-code: falciparum row (dangerous) = RED,

The "hidden" 3 marks: RBC-INVADING PREFERENCE + HYPNOZOITE rule + one RBC-signature

4-mark mini: 5 species + 1 fever word each | 4.8 = add 1 control line + 1 relapse rule + 1 RBC signature

GUT: "malariae = daisy/rosette + OLD RBC" vs "vivax = dots + YOUNG RBC" - the pair the

MNEMONICS + EXAM GUTS

"5 P's": falciparum (crescent) / vivax (dots) / ovale (young + pointed) / malariae (rosette + old)

"QTTK" = quartan - tertian - quotidian + falciparum irregular = 4 fever names

EXAM GUT: "tertian = 48h, quartan = 72h, quotidian = 24h" + "vivax & ovale RELAPSE"

"benign vs malignant" = vivax (benign) / falciparum (malignant) - 1 word each = 1 mark

falciparum = "falci = sickle / crescent" - the word itself tells you the shape

Diagnosis: RDT (Rapid Diagnostic Test) = antigen test; thick film = most sensitive = 0.8-mark

Q12 CORE OUTLETS - Structure & Function of PARAMECIUM caudatum (8 marks)

Draw: 1 BIG labelled Paramecium (slipper-shaped) - 8 parts. 8 points = full marks.

Domain: Eukarya > Protozoa (Ciliophora / Infusoria) > Sub-Phylum Oligohymenophorea > Class Colpodea > Order Peniculida > Family Parameciidae > Genus Paramecium

Free-living freshwater CILIATE; slipper-shaped, 50-300 µm; "infusoria" = old name; Ciliophora = the ciliate class

STRUCTURE - THE 8 LABELS

1. PELICULE (pellicle): flexible outer covering (no rigid wall) - shape + gas exchange by diffusion
2. CILIA: thousands of short, body-covering, rhythmic cilia = LOCOMOTION + sweep food to mouth
3. ORAL GROOVE + CYTOPHARYNX ("mouth"): food funnel; food → FOOD VACUOLE; waste → CYTOPYGE
4. TWO NUCLEI (signature!): MACRO (somatic - food/growth) + MICRO (germ - conjugation, encystation)
5. TWO CONTRACTILE VACUOLES (ant. + post.): pump out excess water = OSMOREGULATION
6. FOOD VACUOLE: intracellular digestion (holozoic nutrition) - the "digestive" part
7. ORAL GUT + COLLECTING / RADIAL canals; KINETO / ORGANELLES for coordinated cilia movement
8. CITOPROCT / CYTOPYGE (excretory pore) + mucocyst (defence) at anterior end

LABEL SET to draw: cilia, oral groove, macronucleus, micronucleus, 2 CV, food vacuole, cytopyge
Identify = "many cilia + 2 nuclei + oral groove + 2 CV" = Ciliophora exam trio (any missing = 0)

FUNCTIONS - no organs, one cell

NUTRITION = HOLOZOIC: cilia sweep bacteria/algae → oral groove → food vacuole → lysosomal digestion
RESPIRATION + **EXCRETION** = no organs: O₂ in + CO₂ out by DIFFUSION through the pellicle
OSMOREGULATION = 2 contractile vacuoles (pulsatile) - fresh-water survival (water enters, CV pump out)
LOCOMOTION = cilia (rhythmic, coordinated); can reverse via a "turning" ciliary pattern
RESPIRATION type = AEROBIC; excretory waste = ammonia (through membrane, no kidney)
IRRITABILITY = senses food / water / touch (no nerve system - diffuse coordination by cytoplasmic streaming)

REPRODUCTION = binary fission (asexual) + conjugation (sexual, 2 new varieties) + encystation (survival)
VARIETIES (intraspecific): B. aurorae / P. bursaria (hosts Zoochlorella = protosymbiosis)
The TWO NUCLEI: Macro = body + food (somatic); Micro = sex / conjugation (germ, meiosis)
"2 CVs + 2 nuclei + oral groove" = the 6-mark label set examiners check

P. bursaria = "symbiotic paramecium" (green algae = photosynthesis food) = protosymbiosis

SYNONYMS (1/1/1)

Ciliophora = Infusoria / Ciliates | Pellicle = Placella | Cytopyge = Cytoproct / anal pore
Macronucleus = "Big Mac" (somatic) | Micronucleus = germ nucleus | "Paramecium" = slipper-shaped
Encystation = resting HYPNOSPORE / METABOLA (dry-season survival)
Contractile vacuole = pulsatile "water pump" | food vacuole = digesting "stomach"
GOTCHA: "Encystation is a RESTING stage, NOT reproduction" - a very tested trap
"Paramecium = ciliate" = cilia for locomotion; the 2-nucleus rule is its identity

DIFFERENCE + HABITAT

vs Amoeba: Amoeba = pseudopod + 1 nucleus; Paramecium = cilia + 2 nuclei + oral groove
vs Euglena: Euglena = flagellum + mixotrophic; Paramecium = cilia + holozoic only
Habitat: stagnant, organic-rich FRESH water; the classic "pond water" lab organism
Size 50-300 µm; "caudatum" = "boot / slipper-shaped" (the species epithet!)
IDENTIFY: "slipper + cilia + 2 nuclei + oral groove + 2 CV" = Paramecium caudatum
P. caudatum = the textbook species; P. aurelia + P. bursaria = other exam varieties

8 MARK = 1 big labelled Paramecium (3) + 5 structure parts each with function (4) + 1 identification line (1)

Draw the slipper shape + 8 labelled pointers; label the 2 nuclei separately (Macro bigger, Micro smaller)

MNEMONICS + EXAM GUTS

Structure "C-O-M-M-C-F" = Cilia / Oral groove / Macro-nucleus / Micro-nucleus / 2 Contractile vacuoles / Food vacuole

EXAM GUT: "TWO NUCLEI" = Macro = Management (food/growth), Micro = Meiosis (conjugation) - the #1 checked point

Q13 CORE OUTLETS - LIFE CYCLE OF PARAMECIUM (Fission + Conjugation + Encystation) (8 marks)

Draw: 4-box life cycle - Fission + Conjugation + Encystation + germination. 8 points = full marks.

F (ASEXUAL) - BINARY FISSION (main)

1. The 2 nuclei divide: MACRO by amitosis, MICRO by meiosis + reformation; body divides LO
2. Result = 2 IDENTICAL daughters = rapid population growth in a healthy pond
3. Trigger: good food + favorable conditions; fastest way to multiply
4. Each daughter = 1 Macro + 1 Micro nucleus + 2 CV + oral groove
5. Asexual; no genetic mixing; fast but low variation

"F = more cells" - it multiplies the population (2 in = 2 out clones)

C (SEXUAL) - CONJUGATION (2 new varieties)

1. 2 cells meet + align (cilia + oral groove); each micronucleus MEOTICALLY splits (4 haploid, 1
2. One haploid = MIGRANT (moves out) + one = RESIDENT (stays); they exchange reciprocally
3. New diploid MACRONUCLEUS forms; the old one degrades; pair separates → 2 exconjugants
4. Trigger: aging / starvation; "CONJUGATING VARIETIES" term = 2 genetically different cells
5. Restores genetic variation in a clone population - the exam "why" line

"C = mix, NOT more" - 2 in = 2 NEW-VARIETY cells out (recombination, not multiplication)

E - ENCYSTATION (resting stage)

1. In dry / starving water: cilia retract + a thick coat forms → resting HYPNOSPORE / METABOL
2. When water returns: the cyst DORMANT BREAKS (germinates) → active ciliate
3. Cyst = resistant to desiccation + low temp; the "spore" of the ciliate
4. Trigger: drying pond / bad food; reversible (germination when conditions improve)
5. Hypnospore = encysted cell; on germination = 1 active paramecium (not 2)

"E = rest, NOT more" - encystation is SURVIVAL, not reproduction (a tested trap)

WHY + Symbiosis + Varieties

WHY conjugation: it restores GENETIC VARIATION in a mostly clonal population; 2 new varieties
TRIGGERS: fission in good food; conjugation when cells are OLD / hungry; encystation when
SYMBIOSIS: P. bursaria hosts green algae (Zoochlorella) = PROTOSYMBIOSIS (cell moves, algae
VARIETIES: B. aurorae (freshwater, 4 micronuclei) vs P. caudatum (freshwater, 1 Micro)
"Exconjugant" = conjugation cell before it divides; "conjugating varieties" = genetically different
Paramecium = the textbook model: shows asexual + sexual + resting all in ONE organism

MNEMONIC "F-C-E" = Fission (more cells) / Conjugation (mix, NOT more) / Encystation (rest, NOT more)

Conjugation = 2 cells in = 2 NEW-VARIETY cells out | Encystation = 1 cell → 1 cyst → 1 cell (survival, not increase)

4-box life cycle diagram (3) + 3 processes, each with role (4) + "encystation ≠ fission"

Draw 4 boxes in a row: Fission (1→2 clones) + Conjugation (2 cells swap haploid) + Encystation (1 cell → 1 cyst) + Germination (1 cyst → 1 cell)
Label the 2 nuclei inside each box; show "2 in = 2 out" for conjugation (NOT 4)

The 2 mark-generators: "encystation is a resting stage, NOT reproduction" + "conjugation produces new varieties, not daughter cells"

4-mark mini: F + C + E, one line each + 1 "2 in = 2 out" | 4.8 = add 1 trigger + 1 variety (B. aurorae)
"Exconjugant" + "conjugating varieties" + "hypnospore / metabola" = 3 extra bonus words (3 marks)

MNEMONICS + EXAM GUTS

"F-C-E" = Fission / Conjugation / Encystation = the 3 life-cycle events (write it first)
Fission = "multiplies" | Conjugation = "mixes genes" | Encystation = "rests" = the 3 meanings
Conjugation 4 stages: 1-met + 2-exchange (migrant in / out) + 3-degrade old Macro + 4-divide
EXAM GUT: "conjugation produces NEW VARIETIES, not daughter cells" + "encystation = resting stage, not reproduction"
"Meiosis = Micro nucleus only" = the mark that shows you know which nucleus divides how
"Aging triggers conjugation" = why the cell mixes genes when it is old / hungry
"Ciliophora has 2 kinds of division: amitosis (Macro) + meiosis (Micro)" = the nucleus rule
Draw: 4-box cycle + show "2 in = 2 out" on conjugation + "1 cyst = 1 cell" on encystation

Q14 CORE OUTLETS - Phylum ANNELIDA: characteristics + 3 classes (8 marks)

Draw: 1 labelled earthworm. 6 phylum characteristics + 3 classes with examples = full marks.

6 PHYLUM CHARACTERISTICS (write each)

1. METAMERISM: body in repeated similar SEGMENTS (metameres); HOMOMERIC (each ring = "Annelida" = "annulus" = small ring (each segment); ~16,500 species)
 2. TRUE COELEM (SCHIZOCOEL): mesoderm-lined coelom filled with coelomic fluid = HYDROSTATIC SKELETON
 3. TRIPOBLASTIC + BILATERAL + PROTOSTOME: 3 germ layers; symmetric; blastopore becomes mouth
 4. CIRCULATION = CLOSED + aortic arches; blood is RED (chlorocruorin + Hb)
 5. EXCRETION by NEPHRIDIA (metanephridia): 1 pair / segment; NERVOUS = ventral cord + subpharyngeal ganglia
 6. RESPIRATION = body surface (moist); locomotion by SETAE + hydrostatic pressure (peristaltic)
- Body: prostomium (snout) + segments + setae (bristles) + clitellum (waist band, earthworm)
 - Ex: Pheretima (earthworm), Nereis (polychaete), Hirudo (leech), Sandworm
 - "Double metamer" = 1 somatic + 1 genital segment pair

THE 3 CLASSES "P-O-H" + examples

1. POLYCHAETA (many bristles) - MARINE
PARAPODIA (fleshy lobes with bristles) + MANY setae; SEPARATE sexes; TROCHOPHORE larva; e.g. Nereis
 2. OLIGOCHAETA (few bristles) - SOIL / fresh water
Few setae, NO parapodia; HERMAPHRODITE (but cross-fertilization + clitellum cocoon); e.g. Lumbricus
 3. HIRUDINEA (leeches) - freshwater / parasitic
Few / NO setae, 2 SUCKERS (anterior + posterior), FIXED 21 segments; e.g. Hirudo, Glossiphonia
- CLASS ID: parapodia = polychaete | suckers + 21 seg = leech | setae only = oligochaete**
 Polychaete = "many + parapodia + separate + larva" | Oligochaete = "few + clitellum + hermaphrodite"
 Hirudinea = "sucker + no setae + 21 segments (fixed, metamerism lost)"
 Class mnemonic "P-O-H" = Poly-chaeta / Oligo-chaeta / Hirudinea

REPRODUCTION (per class)

POLYCHAETES: separate sexes + TROCHOPHORE larva (direct development or planktonic larva)
 OLIGOCHAETES (earthworms) + HIRUDINEA (leeches): HERMAPHRODITES; CROSS-fertilization
 Spermatheca = stores the partner's sperm; clitellum = egg + sperm sac (the cocoon-secreting region)
"No self-fertilization in annelids" = the exam mark (cross only, even when hermaphrodite)
 Polychaete = broadcast + larva; Oligo/Hirudo = cocoon (biparental cross)
 "Parthenogenesis" is rare/absent in annelids - mostly sexual or hermaphrodite cross

SYNONYMS + EXAM GUTS

Metamerism = segmentation | Metamer = segment | Schizocoel = split coelom (true coelom)
 Hydrostatic skeleton = coelomic fluid + setae + circular/long muscles (no bones)
 Protostome = blastopore → mouth; triploblast = 3 germ layers (ecto + meso + endo)
 "Annelida = the segmented worms"; "Oligochaete = few bristles = soil" = 2 signature lines
GUT: "metamerism + true coelom + triploblastic + protostome + closed circulation + nephridia" = 6 marks
 Class "P-O-H": parapodia / suckers / setae-only = the 3-way distinguish

8 MARK = 1 earthworm diagram (3) + 6 phylum characteristics (3) + 3 classes with examples (2)

The 6 characteristic lines: metamerism / true coelom / triploblastic + protostome / closed circulation / nephridia + ventral cord / setae

MNEMONICS + EXAM GUTS

Phylum "M-T-T-C-N-S" = Metamerism / True coelom / Triploblastic+Protostome / Closed circulation / Nephridia / Setae
 Class "P-O-H" = Poly-chaeta (marine, parapodia) / Oligo-chaeta (soil, setae) / Hirudinea (leech, suckers + 21 seg)
 "Annelida = annulus (small ring)" = the etymology mark; "coelom = hydrostatic skeleton" = the "why no bones" mark
EXAM GUT: 6 characteristics (one line each) + 3 classes + 1 example each + "no self-fertilization" = full 8
 4-mark mini: 4 characteristics + 3 classes named | 4.8 = all 6 characteristics + all 3 classes with examples + 1 reproduction line
 Draw: earthworm external (segments + clitellum + setae + prostomium) = the go-to exam diagram

Q15 CORE OUTLETS - EXTERNAL STRUCTURE of Earthworm (Pheretima) (8 marks)

Draw: 1 labelled Pheretima (slightly curved). 8 body parts + 4 "number" facts = full marks.

BODY PARTS (write 5, each 1-2 lines)

1. GENERAL: long cylindrical worm of ~100-150 SEGMENTS; slightly curved; blunt rounded ANTERIOR
 2. PROSTOMIUM: small lobed "snout" over the mouth (before segment 1) - sensory
 3. SEGMENTS / METAMERES: each ring bears 4 bundles of SETAE (bristles) = anchoring for burrowing
 4. CLITELLUM: prominent saddle band on SEGMENTS 34-37; secretes the COCOON (eggs + sperm)
 5. MOUTH + PHARYNX: anterior, sucking pharynx (pulls in soil); posterior = ANUS
- GENITAL PORES: 2 male + 2 female in the genital band (~ segments 14-27)

The 4 "number" FACTS: clitellum = 34-37 | aortic arches ("hearts") = 7-11 | ~100-150 segments | genital band = 14-27

Setae = 4 bundles / segment; clitellum wider + no setae (its own mark)

Colour: reddish-brown; moist body surface (respiration); prostomium = "over-mouth" snout

FUNCTIONS + 4 IDENTIFYING FACTS

Pheretima = OLIGOCHAETA (few setae, no parapodia, hermaphrodite) = the soil earthworm

SETAE = anchors + sense soil (tactile); body moves by peristalsis (circular + longitudinal muscles)

FACT 1: CLITELLUM on segments 34-37 (the wider "waist") - the #1 tested number

FACT 2: AORTIC ARCHES (4 pairs = "hearts") on segments 7-11 - the #2 tested number

FACT 3: ~100-150 segments total; genital band = segments 14-27 (2 male + 2 female pores)

FACT 4: HERMAPHRODITE but CROSS-fertilization only (no self-fertilization) + clitellum cocoons

"Clitellum 34-37 + aortic arches 7-11" = the 2 number-facts they test most

Setae = bristle anchors (absent on clitellum + prostomium); "no parapodia = not a polychaete"

Mouth + sucking pharynx = eats soil + humus; casts (excreted soil) = the fertilizer mark

SYNONYMS + PHERETIMA PROFILE

Pheretima / Earthworm = Oligochaeta | Clitellum = cocoon-sac / genital band | Setae = bristles

Metamerism = segmentation; "soil engineer" = aerator + castings = the 2 economic marks

Pheretima posthuma = the classic NEB / Pioneer earthworm (also "P. maxima")

Respiration: NO gills / lungs - gas exchange across the MOIST body surface (dry soil kills it)

Sense: light + touch (photoreceptors in skin, NO eyes) = the "no eyes" line

IDENTIFY: "clitellum 34-37 + aortic arches 7-11 + ~100-150 seg + setae, no parapodia"

DIFFERENCE FROM OTHER CLASS MATES

vs Nereis (polychaete): Pheretima has NO parapodia + no trochophore larva + hermaphrodite

vs Hirudo (leech): Pheretima has SETAE + NO 2 suckers + many more than 21 segments

"Setae only (no parapodia)" = Oligochaeta signature; "2 suckers + 21 seg" = Hirudinea

Pheretima = free-living, burrowing (aerator); Hirudo = often PARASITIC (blood-sucker)

GUT: "Pheretima = Oligochaeta, setae, clitellum, no suckers" vs "Hirudo = Hirudinea, suckers"

"aortic arches = earthworm hearts" = the circulation mark word

8 MARK = 1 labelled earthworm diagram (3) + 5 body-part descriptions (3) + 4 identifying "number" facts (2)

Label all 8 external parts; the 4 number-facts = clitellum 34-37 / aortic arches 7-11 / ~100-150 seg / genital band 14-27

MNEMONICS + EXAM GUTS

Body parts "P-S-C-M" = Prostomium / Segments (+setae) / Clitellum / Mouth + anus

"4 NUMBER facts": Clitellum = 34-37 | Aortic arches = 7-11 | Segments = 100-150 | Genital band = 14-27 (write all 4)

EXAM GUT: the 2 number-facts "clitellum 34-37 + aortic arches 7-11" are the ones they test hardest

"Setae = anchors" + "no parapodia = oligochaete" + "hermaphrodite but no self-fertilization" = 3 extra marks

Draw: 1 curved earthworm + 8 labelled pointers (prosto / seg / setae / clitellum / mouth / anus / pores / coelom)

Q16 CORE OUTLETS - Metamerism + INTERNAL ORGANIZATION of Earthworm (8 marks)

Draw: 1 internal-organ diagram (gut + aortic arches + nerve cord + nephridia). Define metamerism first.

METAMERISM (define BEFORE the systems)

SEGMENTATION where EXTERNAL body rings AND INTERNAL organs repeat; earthworms = H
 "DOUBLE METAMER" = a somatic + a genital segment pair; metamer = segment = ring
 Coelom = split (schizocoel); each metamer = coelomic compartment + nephridia + nerve gang
GUT: define metamerism FIRST, then describe the 4 systems = the 8-mark order
 Homomeric = similar segments (annelids); Heteromeric = different (arthropods - e.g. head/th
 Metamerism "benefit" = one segment can fail without killing the whole animal (exam line)

THE 4 SYSTEMS (2 lines each)

DIGESTIVE: mouth → pharynx (sucking) → oesophagus → crop + GIZZARD (grinds soil) → int
 CIRCULATORY (CLOSED): dorsal + ventral + setal vessels; blood pushed by AORIC ARCHES ("h
 EXCRETORY (NEPHRIDIA): one pair METANEPHRIDIA / segment (~100+ total); each = flame-cu
 NERVOUS: supra + sub-oesophageal ganglion ("brain") + circum-oesophageal bridge; one v
4 systems = digestive + closed circulatory + nephridia + ventral nerve cord = the internal
 Calcium-cells (in crop) store CaCO_3 - the "lime" mark

REPRODUCTIVE + RESPIRATION + SENSE

HERMAPHRODITE: two testis-cysts + two pair ovaries; CLITELLUM (seg 34-37) builds the COCC
 RESPIRATION: no gills / lungs - gas exchange across the MOIST body surface (why dry soil kill
 SENSE: light + touch (photoreceptors in skin, NO eyes); setae = tactile (soil-reading) sense
"Moist surface = respiration; no eyes; setae = touch" = 3 sensory/respiratory lines
 Coelom = hydrostatic skeleton = the "why no bones" mark
 Excretion = ammonia (diffusion) + coelomic fluid filtered by nephridia

SYNONYMS + EXAM GUTS

Metamerism = segmentation | Metamer = segment | Aortic arches = "earthworm hearts"
 Nephridia = kidney-equivalent | Flame-cup = nephrostome (funnel); metanephridium = 1 se
 Coelom = true body cavity | Hydrostatic skeleton = coelomic fluid + circular/long muscles +
 Intestinal caecum = absorption organ; GIZZARD = grinding; CROP = storage
GUT: "metamerism + coelom + closed circulation + metanephridia + ventral cord + herma
 "Soil engineer = aerator + castings (plant food)" = the significance mark

8 MARK = 1 internal-organ diagram (3) + 4 systems, each 2 lines (4) + 1 metamerism + significance line (1)

Order: metamerism DEFINITION first, then digestive → circulatory → excretory → nervous → reproduc + resp + sense

MNEMONICS + EXAM GUTS (all 4 systems + extra marks)

Metamerism "homomeric = similar; heteromeric = different" - write BOTH types (exam wants the distinction)
 4 systems "D-C-E-N" = Digestive / Closed-Circulatory / Excretory (nephridia) / Nervous (ventral cord)
 Digestive "M-P-O-C-G-I-A" = Mouth-Pharynx-Oesophagus-Crop-Gizzard-Intestine-rectum-Anus (the gut path)
 Circulatory: "aortic arches = hearts, seg 7-11; no WBC (coelomic phagocytes); blood = red (chlorocruorin)"
 Excretory: "flame-cup + coiled tubule + nephridiopore; one pair / segment (~100+ total) = metanephridia"
 Nervous: "brain (supra + sub) + circum-oesophageal bridge + ventral cord + perivisceral nerves; giant fibres = fast escape"
EXAM GUT: "metamerism must be defined BEFORE the systems" + "6 internal marks: coelom + closed circ + nephridia + ventral cord + hermaphrodite cross + moist-surface respiration"
 "No gills / lungs = moist surface; no eyes = photoreceptors; setae = touch" = the 3 sense/respiratory lines
 "Soil engineer = aerator + castings = plant food" = the significance mark; "metamerism = one segment can fail without killing the whole" = the 'why' line
 4.8 extension: add "double metamer (somatic + genital)" + "calcium-cells in crop store CaCO_3 " for the last 0.8